

Path-integral measure and anomalies for left-handed fermions

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A path-integral measure for left-handed fermions is constructed for non-Abelian gauge theories, and its properties under unitary transformations are discussed.

Recently Witten¹ has shown a mathematical inconsistency in an SU(2) gauge theory with an odd number of left-handed fermion doublets. The problem discovered by Witten is somewhat analogous to the Adler-Bell-Jackiw² anomaly and, in his approach, based on the path integral, appears as an ambiguity in the square root of the (left-handed) fermion determinant.

It is the purpose of this note to discuss the integration measure for left-handed fermions showing that problems also related to axial anomalies appear even at earlier stages than those studied by Witten. An analysis of these difficulties within the path-integral framework is presented as well as the conditions under which the invariances of the theory are preserved.

We consider a gauge theory with the (Euclidean) Lagrangian

$$\mathcal{L} = \frac{1}{4} F_{\mu\nu}^2 - \bar{\psi} \mathcal{D} \psi, \quad (1)$$

where the Hermitian operator $\mathcal{D}$ is

$$\mathcal{D} = \gamma_\mu (i \partial_\mu + A_\mu) \quad (2)$$

and A_μ is a gauge field corresponding to some gauge group G .

In order to define the path-integral measure for left-handed fermions, that is, fermions satisfying

$$\gamma_5 \psi = -\psi, \quad \bar{\psi} \gamma_5 = \bar{\psi}, \quad (3)$$

we proceed³ by considering the eigenvalue equation

$$\mathcal{D} \varphi_\lambda = \lambda \varphi_\lambda, \quad \int \varphi_\lambda^\dagger \varphi_{\lambda'} dx = \delta_{\lambda\lambda'}. \quad (4)$$

Now, for any positive eigenvalue λ there is a negative one $-\lambda$,

$$\mathcal{D}(\gamma_5 \varphi_\lambda) = -\lambda \gamma_5 \varphi_\lambda. \quad (5)$$

We shall then take as a basis the set⁴

$$\{\varphi_\lambda, \varphi_{-\lambda} = \gamma_5 \varphi_\lambda\}, \quad \lambda > 0 \quad (6)$$

and expand the classical fields $\psi, \bar{\psi}$:

$$\begin{aligned} \psi &= \sum_{\lambda > 0} (a_\lambda \varphi_\lambda + a_{-\lambda} \varphi_{-\lambda}), \\ \bar{\psi} &= \sum_{\lambda > 0} (\varphi_\lambda^\dagger \bar{a}_\lambda + \varphi_{-\lambda}^\dagger \bar{a}_{-\lambda}), \end{aligned} \quad (7)$$

where $a_{\pm\lambda}, \bar{a}_{\pm\lambda}$ are elements of an infinite Grassmann algebra at the classical level. If one were to integrate over ordinary (Dirac) fields, the functional integral measure should read

$$\mathcal{D}\bar{\psi} \mathcal{D}\psi = \prod_{\lambda > 0} d\bar{a}_{-\lambda} d\bar{a}_\lambda da_{-\lambda} da_\lambda. \quad (8)$$

In order to restrict the measure to left-handed (Weyl) fermions, constraints among the Grassmann variables have to be introduced. To determine these constraints, let us note that for left-handed fermions, the number of independent Grassmann coefficients in an expansion like (7) is one-half of that corresponding to Dirac fermions. Indeed, in order to satisfy condition (3) one has to demand

$$\sum_{\lambda > 0} (\gamma_5 a_\lambda + a_{-\lambda}) \varphi_\lambda = - \sum_{\lambda > 0} (a_\lambda + \gamma_5 a_{-\lambda}) \varphi_\lambda \quad (9)$$

which is automatically verified if the coefficients in the expansion (7) are chosen so that

$$a_\lambda = -a_{-\lambda}. \quad (10)$$

Here the identity means that the generators multiplying the eigenfunctions φ_λ and $\varphi_{-\lambda}$ have to be the same. Analogously, one gets, for $\bar{\psi}$,

$$\bar{a}_\lambda = \bar{a}_{-\lambda}. \quad (11)$$

It is then natural to propose as a left-handed integration measure the expression

$$(\mathcal{D}\bar{\psi}\mathcal{D}\psi)_L = \prod_{\lambda>0} \delta(\bar{a}_\lambda - \bar{a}_{-\lambda}) \delta(a_\lambda + a_{-\lambda}) d\bar{a}_\lambda d\bar{a}_{-\lambda} da_\lambda da_{-\lambda} \quad (12)$$

with

$$\delta(a_i - a_j) = \eta_{ij} (a_i - a_j) \quad , \quad (13)$$

η_{ij} being a sign chosen so to have a consistent definition for the δ function (see, for example, Ref. 5).

We are now ready to consider the following path integral:

$$Z_L = \int (\mathcal{D}\bar{\psi}\mathcal{D}\psi)_L \exp \left[\int \bar{\psi} \mathcal{D}(A) \psi dx \right] \quad . \quad (14)$$

Using the above definitions we have

$$Z_L = \int \prod_{\lambda>0} \delta(\bar{a}_\lambda - \bar{a}_{-\lambda}) d\bar{a}_{-\lambda} d\bar{a}_\lambda \delta(a_\lambda + a_{-\lambda}) da_\lambda da_{-\lambda} \exp \left[\sum_{\lambda, \lambda'>0} \varphi_\lambda^\dagger (\bar{a}_\lambda + \gamma_5 \bar{a}_{-\lambda}) \mathcal{D}(a_{\lambda'} + \gamma_5 a_{-\lambda'}) \varphi_{\lambda'} dx \right] \quad (15)$$

or

$$Z_L = \int \prod_{\lambda>0} d\bar{a}_\lambda da_\lambda \exp \left[\sum_{\lambda>0} 2\lambda \bar{a}_\lambda a_\lambda \right] = \prod_{\lambda>0} (2\lambda) \quad . \quad (16)$$

Hence, apart from a numerical factor (which can be absorbed in the normalization of Z_L) we have obtained an expected result: the restriction of the integration domain to left-handed fermions yields for Z_L the product of only one-half of the eigenvalues, the positive ones. We note that we have obtained the product of *positive* eigenvalues as a result of our choice of the integrated variables (the $a_{-\lambda}, \bar{a}_{-\lambda}$) as well as the δ -function convention. Any other choice will give the same result apart from an overall sign and corresponds to a choice in the sign of the square root [see Eq. (17)].

From Eq. (16) we can *formally* write

$$\int (\mathcal{D}\bar{\psi}\mathcal{D}\psi)_L \exp \left[\int \bar{\psi} \mathcal{D}(A) \psi dx \right] \equiv + [\det \mathcal{D}(A)]^{1/2} \quad . \quad (17)$$

The inclusion of zero modes does not modify expressions (16) and (17) in the sense that in that case $[\det \mathcal{D}(A)]^{1/2} = 0$.

Of course since the eigenvalues of $\mathcal{D}$ increase without bound, in order to define $\prod_{\lambda>0} \lambda$ one has to adopt some regularization procedure. It was recently understood that the use of gauge-invariant regularization methods allows the obtaining of the axial anomaly in the path-integral framework as a result of the Jacobian of a chiral change in the fermionic variables.⁶⁻⁸ For Dirac fermions the integration measure is naturally gauge invariant (actually, this measure is invariant under any unitary transformation $\psi' = \Omega \psi$, $\bar{\psi}' = \bar{\psi} \Omega^{-1}$ which corresponds to a change of basis in the functional space),

but, as was shown by Fujikawa⁶ it has an “anomalous” behavior under chiral changes, precisely yielding the Adler-Bell-Jackiw anomaly.

On the other hand, for left-handed fermions, a unitary transformation

$$\begin{aligned} \psi_L &\rightarrow e^{i\epsilon(x)} \psi_L = \psi_L^\xi \quad , \\ \bar{\psi}_L &\rightarrow \bar{\psi}_L e^{-i\epsilon(x)} = \bar{\psi}_L^\xi \end{aligned} \quad (18)$$

is indistinguishable from the chiral change

$$\begin{aligned} \psi_L &\rightarrow e^{-i\gamma_5 \epsilon(x)} \psi_L = \psi_L^\xi \quad , \\ \bar{\psi}_L &\rightarrow \bar{\psi}_L e^{-i\gamma_5 \epsilon(x)} = \bar{\psi}_L^\xi \quad , \end{aligned} \quad (19)$$

where we shall consider the cases where $\epsilon(x)$ takes values in the Lie algebra of the gauge group G and also when it corresponds to $U(1)$ transformations. Because of this relation, it is not straightforward to define a left-handed measure automatically invariant under unitary transformations. In fact, it is easy to see that, if one considers (18) as an infinitesimal change of variables and uses the measure (12) [which led to Eq. (17)], one then has

$$(\mathcal{D}\bar{\psi}\mathcal{D}\psi)_L = J_L (\mathcal{D}\bar{\psi}^\xi \mathcal{D}\psi^\xi)_L \quad (20)$$

with J_L given by

$$\begin{aligned} J_L &= \exp(-\text{tr} i \gamma_5 \epsilon) \\ &= \exp \left[-2i \sum_{\lambda>0} \int dx \varphi_\lambda^\dagger(x) \gamma_5 \varphi_\lambda(x) \epsilon(x) \right] \end{aligned} \quad (21)$$

[Eq. (21) follows exactly as for the case of *chiral* transformations for Dirac fermions, as was first derived by Fujikawa⁶].

Hence, for left-handed fermions the Jacobian arising from an unitary (infinitesimal) transformation with a measure restricted as in (12), is governed by

the quantity

$$2 \sum_{\lambda>0} \varphi_{\lambda}^{\dagger}(x) \gamma_5 \varphi_{\lambda}(x) = \sum_{\forall \lambda} \varphi_{\lambda}^{\dagger}(x) \gamma_5 \varphi_{\lambda}(x) . \quad (22)$$

Since expression (22) is ill defined, one has to regularize it using, for example, the habitual gauge-invariant procedure of inserting a factor

$$\exp(-\lambda^2/M^2) , \quad (23)$$

and, after making a point separation, studying the limit $M^2 \rightarrow \infty$.

With this technique one obtains

$$J_L = \exp \left[\frac{-i}{16\pi^2} \int \text{tr} F_{\mu\nu} F_{\mu\nu}^* \epsilon(x) dx \right] . \quad (24)$$

Let us analyze the different values of J_L according to the nature of the infinitesimal transformation. We begin by considering that the gauge group is $G=\text{SU}(2)$, with the fermions in the fundamental representation. If the change of variables (18) corresponds to a $\text{U}(1)$ local transformation, then the Jacobian J_L is

$$J_L^{\text{U}(1)} = \exp \left[\frac{-i}{2 \times 16\pi^2} \int F_{\mu\nu}^* F_{\mu\nu}^a \epsilon(x) dx \right] . \quad (25)$$

Hence, the left-handed measure is not invariant

$$\begin{aligned} Z_L(j, \eta, \bar{\eta}) &= \int \mathcal{D}A_{\mu} (\mathcal{D}\bar{\psi} \mathcal{D}\psi)_L \exp \left[- \int (\mathcal{L} + \bar{\eta}\psi + \bar{\psi}\eta + j_{\mu} A_{\mu}) dx \right] , \\ \frac{\delta Z_L}{\delta \epsilon} &= 0 = \int \mathcal{D}A_{\mu} (\mathcal{D}\bar{\psi} \mathcal{D}\psi)_L \exp \left[- \int (\mathcal{L} + \text{source terms}) dx \right] \\ &\quad \times \left[\partial_{\mu} (\bar{\psi} \gamma_{\mu} \psi) + i(\bar{\eta}\psi - \bar{\psi}\eta) + \frac{i}{32\pi^2} T_{\mu\nu}^* F_{\mu\nu}^a \right] , \end{aligned} \quad (29)$$

where the last term is the anomalous part of the identity.

Of course, if, instead of the $\text{U}(1)$ transformation $e^{i\epsilon(x)}$, one performs a *chiral* $\text{U}(1)$ change, the Jacobian is again nontrivial and coincides with (29). For right-handed fermions the Jacobian for $\text{U}(1)$ transformations is the inverse of that obtained for left-handed ones,

$$J_R^{\text{U}(1)} = (J_L^{\text{U}(1)})^{-1} ,$$

while for the case of chiral changes it gives the same result $J_R^{\gamma_5} = J_L^{\gamma_5}$. Hence, if one considers the complete (Dirac) integration domain as the direct sum of left- and right-handed domains, the whole axial anomaly is obtained while the “ $\text{U}(1)$ anomaly” of each sector cancels.

The generalization of the above results to more

under $\text{U}(1)$ transformations. On the contrary, if the fermionic change of variables corresponds to an infinitesimal *gauge* transformation [i.e., $\epsilon(x)$ takes values in the Lie algebra of $\text{SU}(2)$], then in the evaluation of (24) one is led to the computation of

$$\text{tr}([t^a, t^b]_+ t^c) \quad (26)$$

which for the case of $\text{SU}(2)$ generators is zero. Hence

$$J_L^{\text{SU}(2)} = 1 , \quad (27)$$

and we conclude that an $\text{SU}(2)$ gauge theory with left-handed fermions can be consistently handled within the path-integral framework in the sense that the left-handed measure, and hence the generating functional, is gauge invariant. (Of course there is still the problem discovered by Witten for the case in which the number of doublets is odd.) Evidently, the fact that $J_L^{\text{U}(1)}$ is nontrivial implies the existence of an anomalous Ward-Takahashi identity which can be derived by performing a $\text{U}(1)$ transformation like (18) on Z_L and demanding

$$\frac{\delta Z_L}{\delta \epsilon(x)} = 0 . \quad (28)$$

By explicitly including the sources, one has

general gauge groups parallel the usual analysis of γ_5 anomalies in non-Abelian gauge theories.^{9,10} The left-handed measure will be gauge invariant under infinitesimal gauge transformations [apart from the $\text{SU}(2)$ case discussed above] for

$$G = \text{SO}(N), \quad N \geq 5, \neq 6 ,$$

$$G = \text{Sp}(2N), \quad N \geq 3 ,$$

$$G = \text{G}(2) ,$$

$$G = \text{E}(n), \quad n = 4, 7, 8 .$$

For the case of $\text{SU}(3)$, if the representations are $\underline{8}$ or $\underline{3} + \underline{\bar{3}}$, the Jacobian will be trivial. Of course, the $\text{U}(1)$ noninvariance is still present as a reminiscence of the anomaly in the $\text{U}(1)$ axial-vector currents:

$$j_\mu^5 = \bar{\psi} \gamma_\mu \gamma_5 \psi, \quad \partial_\mu j_\mu^5 \neq 0. \quad (30)$$

It is interesting to note that it is not possible to preserve the U(1) invariance by using instead a nonrestricted measure and the projected Lagrangian,

$$\mathcal{L}_{\text{projected}} = \bar{\psi} \mathcal{D}(1 - \gamma_5) \psi \quad (31)$$

since in this case the generating functional is zero [$\mathcal{D}(1 - \gamma_5)$ has an infinite number of zero modes, the right-handed fermions]. Only in the case in which the integration domain is effectively restricted to the left-handed sector do these zero modes not contribute, but, in order to do this one has again to define a left-handed measure of the type (12).

For finite transformations, the use of the methods developed in Refs. 11 and 12 leads to similar results. In particular, for the SU(2) theory, the invariance of the measure under those finite transformations which can be generated by iteration of infinitesimal ones starting from the identity means that

$$[\det \mathcal{D}(A)]^{1/2} = [\det \mathcal{D}(A^g)]^{1/2}, \quad (32)$$

where

$$A^g = g^{-1} A g + i g^{-1} \partial g, \quad g \in \text{SU}(2). \quad (33)$$

Identity (32) can be proven just by considering a change of variables of the form

$$\psi'_L = g \psi_L, \quad \bar{\psi}'_L = \bar{\psi}_L g^{-1} \quad (34)$$

and computing the generating functional before and after than change. In the computation of the Jacobian one uses^{11,12} explicitly that g is constructed from infinitesimal transformations; hence the proof is no longer valid for transformations which, having

nontrivial topological properties, cannot be generated by infinitesimal transformations by starting from the identity. This is precisely the case of the transformation U discussed by Witten¹ for an SU(2) theory. In compactified S^4 space, he considers U 's homotopically nontrivial [this being possible due to the fact that $\pi_4(\text{SU}(2)) = \mathbb{Z}_2$]. For these transformations, using the mod 2 index theorem Witten shows that

$$[\det \mathcal{D}(A)]^{1/2} = -[\det \mathcal{D}(A^U)]^{1/2}. \quad (35)$$

There is no contradiction between Eqs. (32) and (35): identity (32) is valid for transformations constructed from the identity by a series of infinitesimal gauge transformations. On the contrary, Eq. (35) is true for those U 's in SU(2) which cannot be continuously deformed to 1. It may be possible to prove Eq. (35) in our framework by extending the techniques employed to compute the Jacobian to topologically nontrivial gauge transformations, but this may turn out to be very complicated and has not yet been done.

We now conclude by stating the principal point in this investigation: A consistent definition of left-handed path-integral fermionic measure implies, in general, the existence of anomalies under local unitary transformations. However, for certain gauge groups (precisely those where axial anomalies are absent) the left-handed measure is gauge invariant although U(1) anomalies are present as a reminiscence of the U(1) axial vector anomalies.

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